

Integration Report

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Abstract

This integration report includes some additional details regarding some aspects of the project that were not covered in the *Technical Report*.

CCS Concepts

• **Software and its engineering** → **Real-time systems software**; **Embedded software**; *Software architectures*.

Keywords

Real-Time Kernels, Real-Time Systems, Embedded Software, Software Architectures

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1 Why we used a one-to-one mapping between tasks and watchdogs in the Rust System?

Since *RTIC framework* does not provide any built-in mechanism to handle task deadlines, we decided to use a one-to-one mapping between tasks and watchdogs. This way, each watchdog is responsible for monitoring the deadline of a single task, making the implementation simpler and more straightforward.

Besides the simplicity of implementation, this approach was chosen to avoid the runtime cost of managing a pool of deadlines and `deadline_protected_objects` at each deadline check. In fact, the alternative scenario of having a single watchdog monitoring multiple tasks would require to identify over a pool of deadlines, at each deadline check, the earliest next deadline in order to delay its execution accordingly. This operation is needed also every task activation, since it could introduce a new earliest deadline.

Even if we could optimize this operation by using an appropriate data structure, such as a min-heap, the one-to-one mapping approach, instead, allows to manage each watchdog independently, without the need of any search, insert or delete operation over a pool of objects. This results in a constant-time complexity for each deadline check and task activation, thus reducing the runtime overhead, improving the system's responsiveness and reducing

watchdogs' interference over system tasks.

Moreover, since the *RTIC framework* requires to declare for each task locals and shared resources at compile time, by using a one-to-one mapping we can leverage the single responsibility principle, enhancing code readability and maintainability against a more complex single watchdog implementation.

However, this approach introduces a higher activation cost, since each watchdog consists in a task, and as such it has its own activation overhead. This is a price that we considered more acceptable than the previous described runtime cost, since the activation cost is paid only once and does not affect the system responsiveness.

Activation cost is not the only negative aspect of this approach: by having a one-to-one mapping, we also introduce a memory overhead, since each watchdog task requires its own stack and context. This could be a problem in systems with a large number of tasks, where the memory overhead could become significant. In our case, given the limited number of tasks in the system, we considered this overhead acceptable, but we acknowledge that in systems with a larger number of tasks, this could become a trade-off to consider carefully against the runtime cost of a single watchdog implementation.

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